

ASSESSMENT : CompTia A+ - Troubleshooting and Helpdesk
Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installation-verification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

Question 1:

Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.

Answer:

When a newly assembled PC fails to boot, I would first check the **PSU/power connections, RAM, and CPU/motherboard** because these components are essential for the system to complete its initial hardware checks and POST.

1. PSU and Power Connections

I would first check whether the PSU is switched on and receiving power. I would verify that the **24-pin ATX motherboard power connector** and the **CPU EPS power connector** are properly connected. If a dedicated GPU is installed, I would also check its required power connection.

A loose or missing power connection can prevent the system from powering on or completing POST.

2. RAM

Next, I would check the RAM because incorrectly installed or incompatible memory can cause a system to fail POST. I would power off the system, reseal the RAM module and make sure it is properly locked into the memory slot.

If multiple RAM modules are installed, I would test them individually in the recommended slot to identify a possible faulty module or slot.

3. CPU and Motherboard

Finally, I would check the CPU and motherboard. I would verify that the CPU is compatible with the motherboard and that the motherboard BIOS/UEFI supports the processor. I would also check the CPU power connection and ensure that the CPU cooler is correctly installed.

If the motherboard has diagnostic LEDs, POST codes, or beep codes, I would use them to identify whether the CPU or motherboard is reporting a hardware problem.

Conclusion:

The PSU/power connections, RAM, and CPU/motherboard are the first three areas I would inspect because they are fundamental to powering the system and completing POST.

Checking these components systematically can help identify common causes of a no-boot condition before investigating less critical components.

Question 2 :

Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.

Answer

Motherboard–CPU compatibility is essential because the processor must be electrically and physically compatible with the motherboard. The motherboard must support the processor's CPU package, chipset/platform requirements, firmware, power requirements, and other necessary features.

In the inspected system, the computer is an **ASUS TUF Gaming F15 FX506HC** with an **11th Generation Intel Core i5-11400H** processor. ASUS lists the FX506HC configuration with this processor.

The Intel Core i5-11400H is a mobile processor with an **FCBGA1787** package. This means it is soldered directly to the laptop motherboard rather than installed in a removable desktop-style CPU socket.

Therefore, the processor and motherboard in this laptop are designed as a compatible fixed platform. The CPU cannot normally be removed and replaced like a desktop CPU.

If an incompatible CPU were used in a desktop system, possible problems could include:

- The CPU may not physically fit the motherboard socket.
- The motherboard may not support the processor.
- The BIOS/UEFI may not recognise the CPU.
- The system may fail to complete POST.
- The computer may fail to boot or may become unstable.
- Power or thermal requirements may not be suitable.

Before assembling a desktop system, I would therefore check the **CPU socket/package, motherboard CPU-support list, BIOS/UEFI support, chipset compatibility, and power/thermal requirements** before installing the processor.

Conclusion:

Motherboard–CPU compatibility directly affects whether a computer can initialise and operate correctly. The inspected ASUS TUF Gaming F15 demonstrates this principle because its i5-11400H is a supported, soldered mobile processor specifically used with the FX506HC platform.

Question 3 :

Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.

Answer:

To verify that the RAM and storage devices are installed correctly, I would first shut down the computer, disconnect the power supply, and follow proper ESD safety precautions.

RAM Verification:

- Check that each RAM module is properly seated in its slot.
- Ensure that the locking clips on the RAM slots are fully engaged.
- Start the computer and enter the BIOS/UEFI setup.
- Confirm that the BIOS detects the correct total RAM capacity.
- In my ASUS TUF Gaming F15 FX506HC, there are **two 8 GB RAM modules, giving a total of 16 GB RAM.**
- The system should complete POST without memory-related error messages, beeps, or diagnostic LED indications.

Storage Verification:

- Physically check that the M.2 SSD is properly inserted into the M.2 slot and secured with its screw.
- Enter BIOS/UEFI and verify that the SSD is detected.
- Check the SSD model and capacity if displayed by the BIOS.
- After booting into Windows, verify the storage device through Disk Management, File Explorer, or Device Manager.
- My laptop has a **512 GB NVMe M.2 SSD**, which Windows reports as approximately **477 GB usable capacity** because of the difference between advertised and operating-system-reported storage capacity.

BIOS/POST Confirmation:

BIOS/UEFI should detect the installed RAM and storage device correctly. POST should complete successfully without any RAM or other hardware-related errors. If a component is not detected, I would shut down the system, reseal the component, check its connection, and test it again.

Question 4 :

Produce a hardware-installation checklist with at least three steps that can help prevent similar boot or hardware-detection failures.

Answer:

The following hardware-installation checklist can help prevent boot failures and hardware-detection problems during workstation assembly:

1. Check Component Compatibility

- Confirm that the CPU, motherboard, RAM, storage, GPU, and PSU are compatible before installation.
- Check the motherboard CPU support list and required BIOS/UEFI version.
- Verify that the RAM type and storage interface are supported by the motherboard.

2. Install and Secure Components Correctly

- Ensure that RAM modules are fully seated and their locking clips are engaged.
- Install the CPU correctly and apply the cooling solution properly.
- Ensure that the M.2 SSD and other storage devices are properly connected and secured.
- Install the GPU correctly in its slot and connect any required power cables.

3. Verify All Power Connections

- Check the motherboard 24-pin ATX power connector.
- Check the CPU/EPS power connector.
- Connect the required GPU and storage power cables.
- Ensure that all connections are firmly seated before powering on the system.

4. Perform a Final Physical Inspection

- Check for loose screws, disconnected cables, damaged connectors, or incorrectly installed components.
- Make sure there are no tools or metal objects left inside the case.
- Ensure that cooling fans are connected and able to operate.

5. Perform BIOS/UEFI and POST Verification

- Power on the system and confirm that POST completes successfully.
- Enter BIOS/UEFI and verify that the CPU, RAM, and storage devices are detected correctly.
- If an error occurs, use diagnostic LEDs, beep codes, or BIOS messages to identify the faulty or incorrectly installed component.

Conclusion:

Following this checklist before and after assembly helps reduce installation errors, compatibility problems, boot failures, and hardware-detection issues.

Question 5 :

Provide a technical justification for choosing an SSD instead of an HDD for modern office workstations, relating your answer to the lab use case.

Answer:

An SSD is more suitable than an HDD for modern office workstations because it provides faster performance, better responsiveness, and greater reliability.

1. Faster Data Access

- SSDs use flash memory and have no moving mechanical parts.
- They provide much faster read and write speeds than traditional HDDs.
- This allows the operating system, applications, and files to load more quickly.

2. Faster Boot and Application Loading

- A workstation with an SSD can boot the operating system faster.
- Office applications such as Microsoft Office, web browsers, development tools, and other productivity software can start more quickly.
- This improves overall user productivity in a computer lab.

3. Better Reliability

- HDDs contain mechanical components such as spinning platters and moving read/write heads.
- SSDs have no moving parts, making them more resistant to physical shocks and vibration.
- This is useful in a busy computer-lab environment where systems may be frequently used and handled.

4. Lower Noise and Power Consumption

- SSDs operate silently because they do not contain spinning mechanical parts.
- They generally consume less power than mechanical HDDs, which can also reduce heat generation.

5. Suitable for the ABC Solutions Lab

- The computer lab will require systems that can start quickly, run multiple office applications smoothly, and provide responsive file access.
- An NVMe SSD is especially suitable because it uses the PCIe interface and can provide higher performance than a traditional SATA HDD.